Class 4 Assignment

July 9, 2026

[2]:
```
#We must first import the Pandas library. We will shortcut it to "pd".
import pandas as pd

# The following code will process the current filepath for the dataset

#This imports the necessary libraries
import os
from os.path import curdir

'''
The os.path.join function is used to merge strings (text) that correspond to
file and directory names together to a viable file path for your Operating
System. I printed the result to give more clarity.
'''
csv_path = os.path.join(curdir,'Data','MMA_860_Grocery_Data.csv')
print(csv_path)
```

```
./Data/MMA_860_Grocery_Data.csv
```

[12]:
```
'''
We repeat the above procedure, substituting the excel file name into the
join arguments and using the read_excel function instead.
'''
xl_path = os.path.join(curdir,'Data',"MMA_860_Grocery_Data.csv")
df_xl = pd.read_csv(xl_path,
                    index_col=0)
df_xl.head()
```

[12]:

Obs	Grocery_Bill	N_Adults	Family_Income	Family_Size	N_Vehicles	Distance_to_Store	Vegetarian	N_Children	Family_Pet
1	$357.73	2	$142,141	4	3				
2	$276.84	2	$145,916	2	1				
3	$197.92	1	$86,185	1	2				
4	$315.75	2	$145,998	3	1				
5	$202.89	1	$79,341	1	2				

```
1                  15           0           2           1
2                   4           0           0           0
3                  14           0           0           0
4                   8           0           1           0
5                  19           1           0           0
```

[14]:
```
df_xl.dtypes
```

[14]:
```
Grocery_Bill           str
N_Adults             int64
Family_Income          str
Family_Size          int64
N_Vehicles           int64
Distance_to_Store    int64
Vegetarian           int64
N_Children           int64
Family_Pet           int64
dtype: object
```

[15]:
```
#First declare a function that removes a dollar sign from a string.
def remove_dollar_sign(s):
    '''
    The Python String replace function can be used to replace any
    instances of a substring in a string with another substring.
    In this case, we are replacing "$" with "" (an empty string);
    this effectively gets rid of any dollar signs in your string.
    '''
    return s.replace("$","")
```

[16]:
```
df_xl["Grocery_Bill"] = df_xl["Grocery_Bill"].apply(remove_dollar_sign)
```

[18]:
```
df_xl.head()
```

[18]:
```
     Grocery_Bill  N_Adults Family_Income  Family_Size  N_Vehicles  \
Obs
1          357.73         2     $142,141            4           3
2          276.84         2     $145,916            2           1
3          197.92         1      $86,185            1           2
4          315.75         2     $145,998            3           1
5          202.89         1      $79,341            1           2

     Distance_to_Store  Vegetarian  N_Children  Family_Pet
Obs
1                   15           0           2           1
2                    4           0           0           0
3                   14           0           0           0
4                    8           0           1           0
```

```
5                  19           1           0           0
```

[19]:
```
df_xl["Family_Income"] = df_xl["Family_Income"].apply(remove_dollar_sign)
```

[20]:
```
df_xl.head()
```

[20]:
```
     Grocery_Bill  N_Adults Family_Income  Family_Size  N_Vehicles  \
Obs
1          357.73         2       142,141            4           3
2          276.84         2       145,916            2           1
3          197.92         1        86,185            1           2
4          315.75         2       145,998            3           1
5          202.89         1        79,341            1           2

     Distance_to_Store  Vegetarian  N_Children  Family_Pet
Obs
1                   15           0           2           1
2                    4           0           0           0
3                   14           0           0           0
4                    8           0           1           0
5                   19           1           0           0
```

[21]:
```
'''
Family_Income contains a leading dollar sign as well as commas. We must
write an additional function to remove the commas. We can reuse the general
format from the dollar sign.
'''
def remove_comma(s):
    return s.replace(",","")
```

[22]:
```
df_xl["Family_Income"] = df_xl["Family_Income"].apply(remove_comma)
```

[23]:
```
df_xl.head()
```

[23]:
```
     Grocery_Bill  N_Adults Family_Income  Family_Size  N_Vehicles  \
Obs
1          357.73         2       142141            4           3
2          276.84         2       145916            2           1
3          197.92         1        86185            1           2
4          315.75         2       145998            3           1
5          202.89         1        79341            1           2

     Distance_to_Store  Vegetarian  N_Children  Family_Pet
Obs
1                   15           0           2           1
2                    4           0           0           0
3                   14           0           0           0
```

```
4                    8           0           1           0
5                   19           1           0           0
```

[24]:
```
df_xl = df_xl.astype({"Family_Income": float, "Grocery_Bill": float})
```

[25]:
```
df_xl.dtypes
```

[25]:
```
Grocery_Bill         float64
N_Adults               int64
Family_Income        float64
Family_Size            int64
N_Vehicles             int64
Distance_to_Store      int64
Vegetarian             int64
N_Children             int64
Family_Pet             int64
dtype: object
```

[26]:
```
import statsmodels.api as sm
from statsmodels.formula.api import ols
```

[29]:
```
model = ols('Grocery_Bill ~ Family_Size + Family_Income + N_Adults + N_Children␣
  ↪+ N_Vehicles + Distance_to_Store + Vegetarian + Family_Pet',df_xl).fit()
print(model.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:           Grocery_Bill   R-squared:                       0.839
Model:                            OLS   Adj. R-squared:                  0.838
Method:                 Least Squares   F-statistic:                     740.1
Date:                Thu, 09 Jul 2026   Prob (F-statistic):               0.00
Time:                        18:33:08   Log-Likelihood:                -4900.2
No. Observations:                1000   AIC:                             9816.
Df Residuals:                     992   BIC:                             9856.
Df Model:                           7
Covariance Type:            nonrobust
==============================================================================
=====
                     coef    std err          t      P>|t|      [0.025
0.975]
------------------------------------------------------------------------------
-----
Intercept           23.3057      5.715      4.078      0.000      12.091
34.520
Family_Size         27.6127      1.277     21.619      0.000      25.106
30.119
Family_Income        0.0009   8.77e-05     10.762      0.000       0.001
0.001
```

```
N_Adults              26.8743      2.637     10.193      0.000      21.700
32.048
N_Children             0.7384      1.608      0.459      0.646      -2.416
3.893
N_Vehicles            -0.8388      1.149     -0.730      0.465      -3.093
1.416
Distance_to_Store      3.8897      0.191     20.391      0.000       3.515
4.264
Vegetarian            -7.8462      4.254     -1.844      0.065     -16.194
0.501
Family_Pet             1.6358      2.827      0.579      0.563      -3.913
7.184
==============================================================================
Omnibus:                       51.552   Durbin-Watson:                   2.096
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              172.112
Skew:                           0.081   Prob(JB):                     4.23e-38
Kurtosis:                       5.026   Cond. No.                     3.67e+20
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly
specified.
[2] The smallest eigenvalue is 8.53e-29. This might indicate that there are
strong multicollinearity problems or that the design matrix is singular.
```

[30]:
```
# Getting rid of vehicles first since bad p value with t stat. Seems not too␣
↪significant

model = ols('Grocery_Bill ~ Family_Size + Family_Income + N_Adults + N_Children␣
↪+ Distance_to_Store + Vegetarian + Family_Pet',df_xl).fit()
print(model.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:           Grocery_Bill   R-squared:                       0.839
Model:                            OLS   Adj. R-squared:                  0.838
Method:                 Least Squares   F-statistic:                     863.7
Date:                Thu, 09 Jul 2026   Prob (F-statistic):               0.00
Time:                        18:37:14   Log-Likelihood:                -4900.4
No. Observations:                1000   AIC:                             9815.
Df Residuals:                     993   BIC:                             9849.
Df Model:                           6
Covariance Type:            nonrobust
==============================================================================
=====
                     coef    std err          t      P>|t|      [0.025
0.975]
------------------------------------------------------------------------------
```

```
-----
Intercept            21.7000      5.273      4.115      0.000      11.352
32.048
Family_Size          27.5776      1.276     21.612      0.000      25.074
30.082
Family_Income         0.0009   8.77e-05     10.778      0.000       0.001
0.001
N_Adults             26.8361      2.635     10.183      0.000      21.664
32.008
N_Children            0.7416      1.607      0.461      0.645      -2.413
3.896
Distance_to_Store     3.8913      0.191     20.405      0.000       3.517
4.266
Vegetarian           -7.7810      4.252     -1.830      0.068     -16.125
0.563
Family_Pet            1.7474      2.823      0.619      0.536      -3.792
7.287
==============================================================================
Omnibus:                       52.124   Durbin-Watson:                   2.097
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              175.903
Skew:                           0.079   Prob(JB):                     6.35e-39
Kurtosis:                       5.049   Cond. No.                     1.78e+20
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly
specified.
[2] The smallest eigenvalue is 3.63e-28. This might indicate that there are
strong multicollinearity problems or that the design matrix is singular.
```

[32]:
```
# Children seems to be bad data as well, however, we will instead remove family␣
 ↪size first since we have breakdown of children and adults and they might␣
 ↪have different spending habits

model = ols('Grocery_Bill ~  Family_Income + N_Adults + N_Children +␣
 ↪Distance_to_Store + Vegetarian + Family_Pet',df_xl).fit()
print(model.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:           Grocery_Bill   R-squared:                       0.839
Model:                            OLS   Adj. R-squared:                  0.838
Method:                 Least Squares   F-statistic:                     863.7
Date:                Thu, 09 Jul 2026   Prob (F-statistic):               0.00
Time:                        18:39:11   Log-Likelihood:                -4900.4
No. Observations:                1000   AIC:                             9815.
Df Residuals:                     993   BIC:                             9849.
Df Model:                           6
```

```
Covariance Type:            nonrobust
==============================================================================
=====
                         coef    std err          t      P>|t|      [0.025
0.975]
------------------------------------------------------------------------------
-----
Intercept             21.7000      5.273      4.115      0.000      11.352
32.048
Family_Income          0.0009   8.77e-05     10.778      0.000       0.001
0.001
N_Adults              54.4137      3.816     14.258      0.000      46.925
61.903
N_Children            28.3192      1.216     23.296      0.000      25.934
30.705
Distance_to_Store      3.8913      0.191     20.405      0.000       3.517
4.266
Vegetarian            -7.7810      4.252     -1.830      0.068     -16.125
0.563
Family_Pet             1.7474      2.823      0.619      0.536      -3.792
7.287
==============================================================================
Omnibus:                       52.124   Durbin-Watson:                   2.097
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              175.903
Skew:                           0.079   Prob(JB):                     6.35e-39
Kurtosis:                       5.049   Cond. No.                     5.81e+05
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly
specified.
[2] The condition number is large, 5.81e+05. This might indicate that there are
strong multicollinearity or other numerical problems.
```

[35]:
```
# Pet does not influence grocery bill. P value with T stat is different

model = ols('Grocery_Bill ~  Family_Income + N_Adults + N_Children +␣
  ↪Distance_to_Store + Vegetarian',df_xl).fit()
print(model.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:           Grocery_Bill   R-squared:                       0.839
Model:                            OLS   Adj. R-squared:                  0.838
Method:                 Least Squares   F-statistic:                     1037.
Date:                Thu, 09 Jul 2026   Prob (F-statistic):               0.00
Time:                        18:41:34   Log-Likelihood:                -4900.6
No. Observations:                1000   AIC:                             9813.
```

```
Df Residuals:                      994   BIC:                             9843.
Df Model:                            5
Covariance Type:            nonrobust
==============================================================================
=====
                         coef    std err          t      P>|t|      [0.025
0.975]
------------------------------------------------------------------------------
-----
Intercept             22.0447      5.242      4.205      0.000      11.758
32.332
Family_Income          0.0009   8.76e-05     10.773      0.000       0.001
0.001
N_Adults              54.4312      3.815     14.267      0.000      46.945
61.918
N_Children            28.3297      1.215     23.314      0.000      25.945
30.714
Distance_to_Store      3.8896      0.191     20.405      0.000       3.516
4.264
Vegetarian            -7.8390      4.250     -1.845      0.065     -16.178
0.500
==============================================================================
Omnibus:                       52.446   Durbin-Watson:                   2.094
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              177.774
Skew:                           0.080   Prob(JB):                     2.49e-39
Kurtosis:                       5.059   Cond. No.                     5.78e+05
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly
specified.
[2] The condition number is large, 5.78e+05. This might indicate that there are
strong multicollinearity or other numerical problems.
```

[ ]: